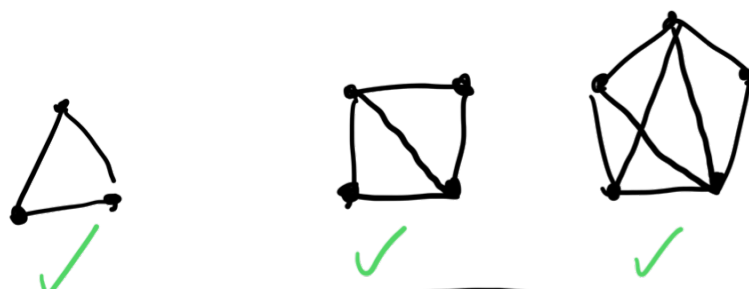


Chordal Graphs

a graph is chordal if every cycle of size 4 or more has a chord (a cross-edge).



Let $\text{HALF-CLQ} = \{ \langle G \rangle \mid G \text{ is a simple, finite graph having a clique subgraph with at least } n/2 \text{ nodes, where } |V| = n. \}$

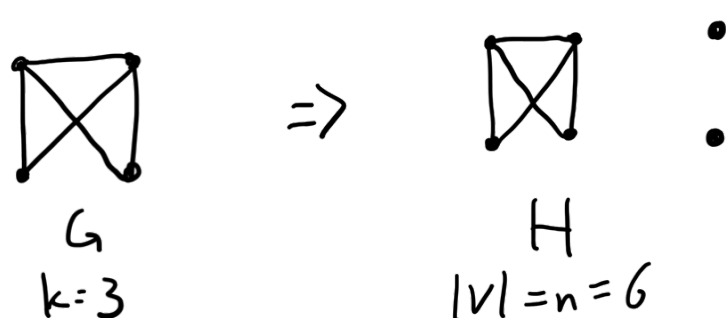


Theorem 7.23. HALF-CLQ is NPC.

Proof: First, we show it is in NP. Given a graph G and a subgraph $S \subseteq G$, it is easy to check whether S is a half-clique by iterating through the vertices of S to see if they are all adjacent (in $O(n)$) and then checking that $|S| \geq |V|/2$.

Next, we show it is NP-Hard via a reduction from CLIQUE. In CLIQUE, we are given k and a graph G and asked if G has a k -clique.

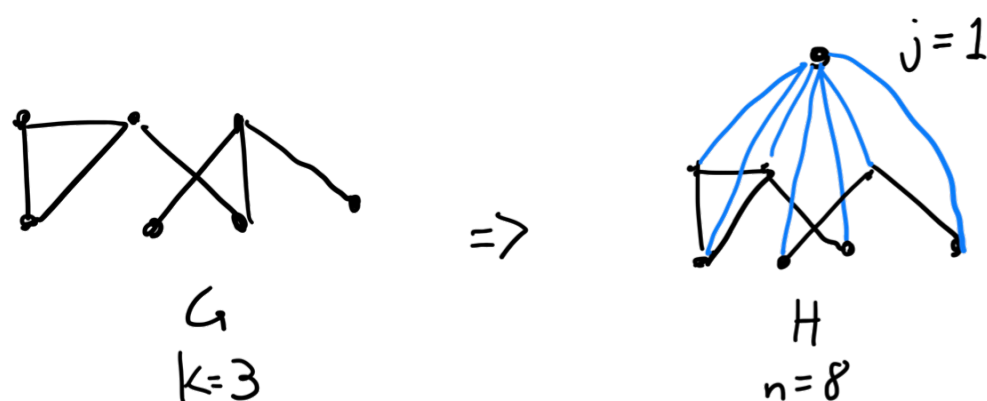
Case 1



If $k \geq n/2$, we add j nodes (an indep. set) to form graph H s.t. H has $2k$ vertices.

Clearly, H has a clique of size $n/2$ iff G has a clique of size k .

Case 2



If $k < n/2$, we add j nodes adjacent to all of G & to one another. s.t. $j = n - 2k$.

$\Rightarrow$ Let G have a clique of size k . Any arbitrary clique in G combines with the j new nodes to make a larger clique. Thus, H has a clique of size exactly $n/2$ (a half-clique).

$\Leftarrow$ Let H have a half-clique. Clearly, H has a k -clique because $k < n/2$. Thus, G has a k -clique.

So, HALF-CLQ is NP-Hard $\cap$ NP. It is NPC.